

Experiment No.	1
Experiment Title	Working with Python Packages: NumPy, SciPy, Scikit-Learn, Matplotlib, and Pandas (Loan Prediction Dataset)
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1 Objective

The core goals of this laboratory exercise comprise:

- Master fundamental data manipulation, array operations, statistical routines, and data visualization primitives across the standard Python scientific stack (NumPy, SciPy, Scikit-Learn, Matplotlib, Pandas).
- Construct a reusable, automated Exploratory Data Analysis (EDA) engine capable of ingesting arbitrary tabular data and extracting statistical parameters, missing value profiles, and distribution visual shapes.
- Categorize five standard benchmark datasets (Iris, Loan Amount Prediction, Diabetes, Email Spam, MNIST) by ML paradigm, target variable characteristics, feature selection methodologies, and model choices.
- Perform full automated EDA on the continuous **Loan Amount Prediction Dataset**, analyzing generated visual outputs and statistical summaries.

2 Problem Statement

Prior to modeling real-world tabular data, data practitioners must perform comprehensive data auditing to identify skewness, unexpected multi-modal distributions, correlation structures, and missing entries. Manual analysis for every newly ingested dataset introduces code redundancy and inefficiency. This experiment addresses the challenge by implementing an end-to-end automated EDA tool using Python's numerical stack to load datasets, generate statistical summaries, and save informative visual representations. Furthermore, standard benchmark datasets are mapped into supervised and unsupervised paradigms to structure future model selection.

3 Dataset Description

The empirical analysis in this section focuses on the ****Loan Amount Prediction Dataset****.

Dataset Name	Loan Amount Prediction Dataset
Dataset Source	UCI Machine Learning Repository / Kaggle
Number of Samples	614
Number of Features	13 (5 continuous/discrete numerical, 8 categorical)

Number of Classes	N/A (Continuous Target: LoanAmount)
Missing Values	Present (Gender, Married, Dependents, Self_Employed, LoanAmount, Loan_Amount_Term, Credit_History)
Train-Test Split	N/A (Exploratory Analysis Phase)

The dataset consists of the following attributes:

- **Loan_ID**: Unique identifier string for each loan application.
- **Gender**: Applicant gender (Male / Female).
- **Married**: Marital status of the applicant (Yes / No).
- **Dependents**: Count of financial dependents (0, 1, 2, 3+).
- **Education**: Educational level (Graduate / Not Graduate).
- **Self_Employed**: Employment type indicator (Yes / No).
- **ApplicantIncome**: Primary applicant monthly income in USD.
- **CoapplicantIncome**: Additional co-applicant monthly income in USD.
- **LoanAmount**: Requested credit amount in thousands of USD (Target).
- **Loan_Amount_Term**: Loan repayment period expressed in months.
- **Credit_History**: Credit guideline compliance indicator (1.0 = positive, 0.0 = negative).
- **Property_Area**: Geographical location of property (Urban / Semiurban / Rural).
- **Loan_Status**: Historical loan approval status (Y / N).

4 Exploratory Data Analysis

The automated EDA framework generated 28 diagnostic graphics saved in the `figures_loan/` directory. Individual figures and corresponding analytical inferences are presented below.

4.1 Histograms of Numerical Features

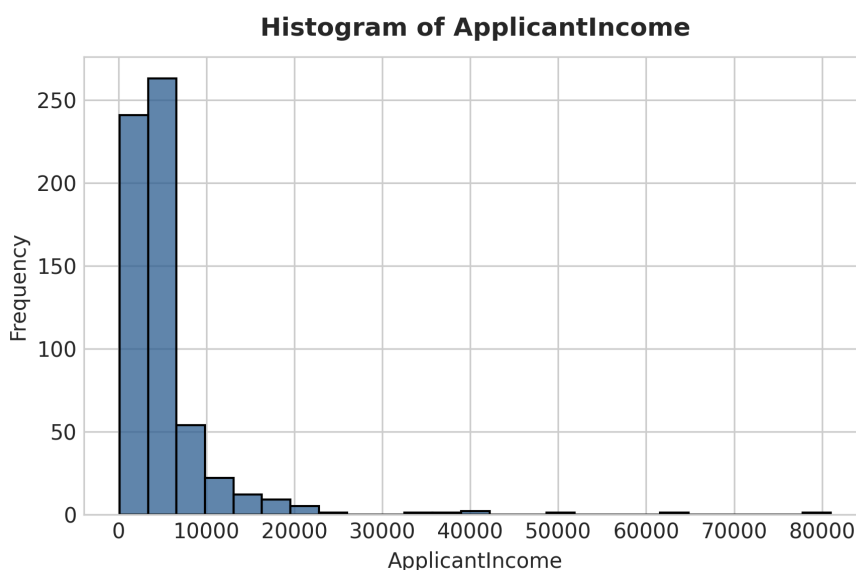


Figure 1: Histogram of Applicant Income

Inference: The applicant income distribution exhibits pronounced right skewness. Most applicants report monthly incomes below \$10,000, while a thin tail reaches toward \$80,000. Applying a logarithmic transformation prior to model fitting is advisable to mitigate variance instability.

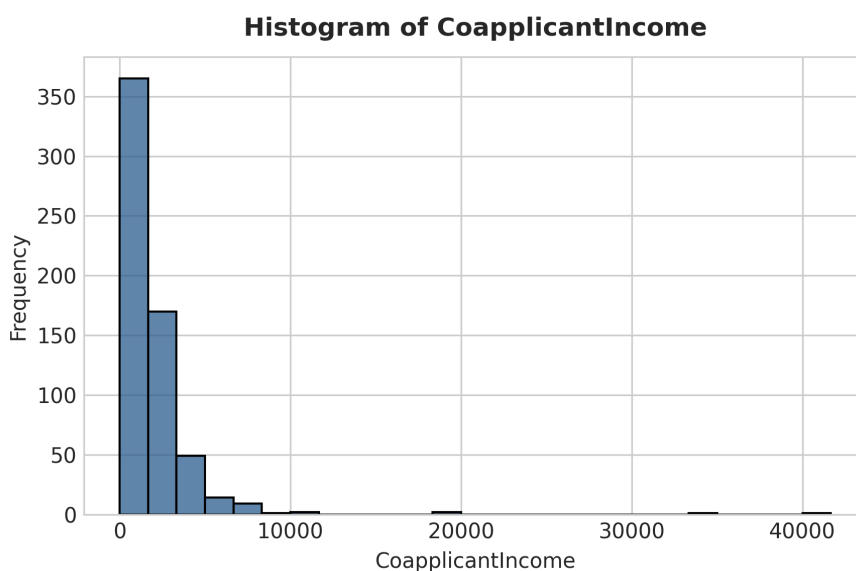


Figure 2: Histogram of Co-applicant Income

Inference: Co-applicant income features a massive spike at \$0, representing applications submitted without a co-borrower. Non-zero entries show positive skewness stretching up to \$40,000.

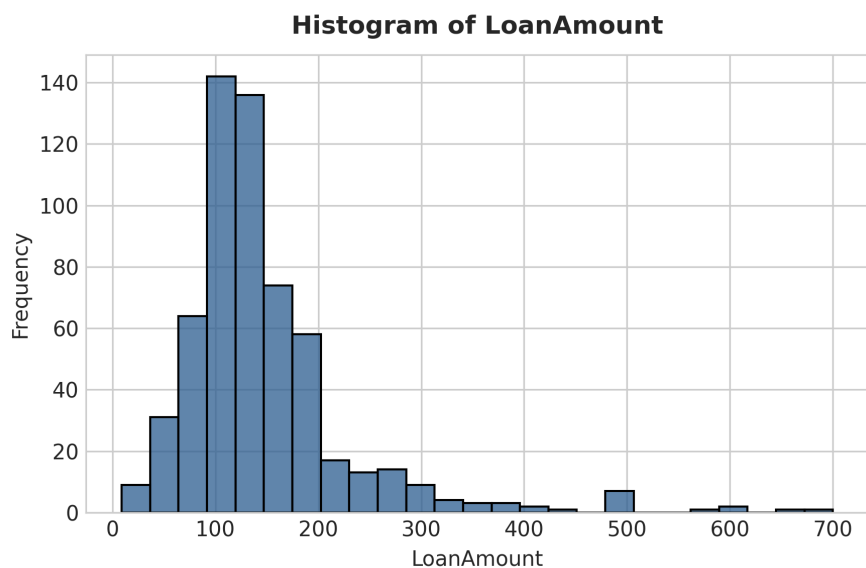


Figure 3: Histogram of Loan Amount

Inference: The target variable (`LoanAmount`) displays a unimodal distribution centered around \$130,000-\$150,000, with minor right-skewness and values spanning \$9,000 to \$700,000.

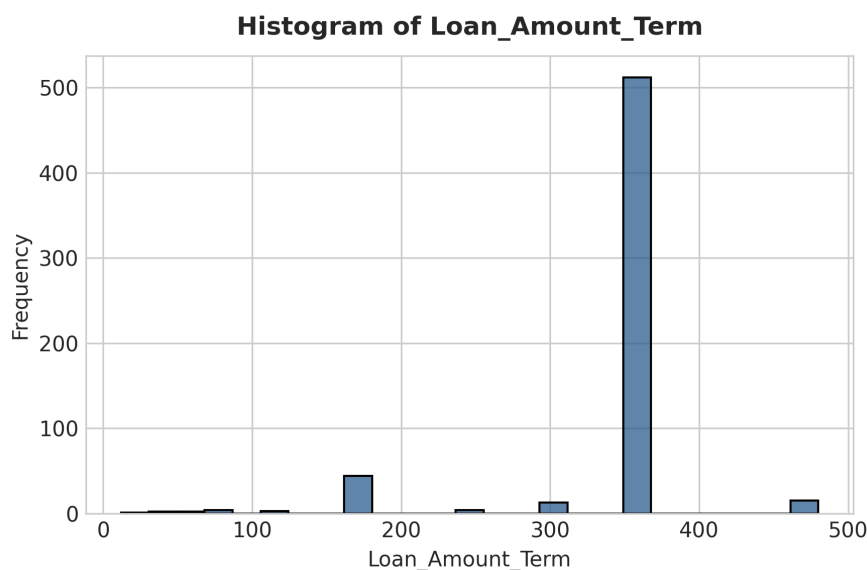


Figure 4: Histogram of Loan Amount Term

Inference: The repayment term distribution is discrete, dominated heavily by the standard 360-month (30-year) mortgage period. Intermediate terms like 180 or 240 months appear far less frequently.

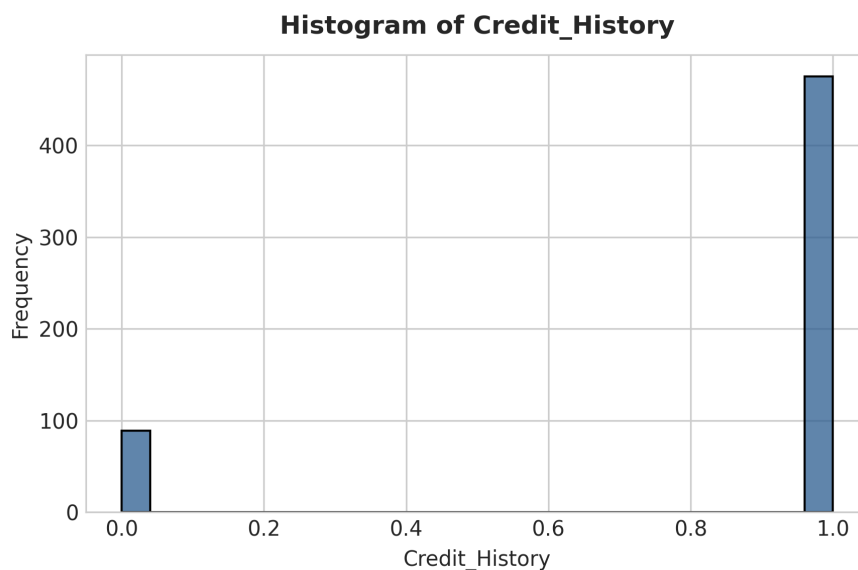


Figure 5: Histogram of Credit History

Inference: Credit history records fall into two binary bins. A significant majority of applicants possess a positive history (`Credit_History` = 1.0), indicating potential class imbalance in this attribute.

4.2 Box Plots of Numerical Features

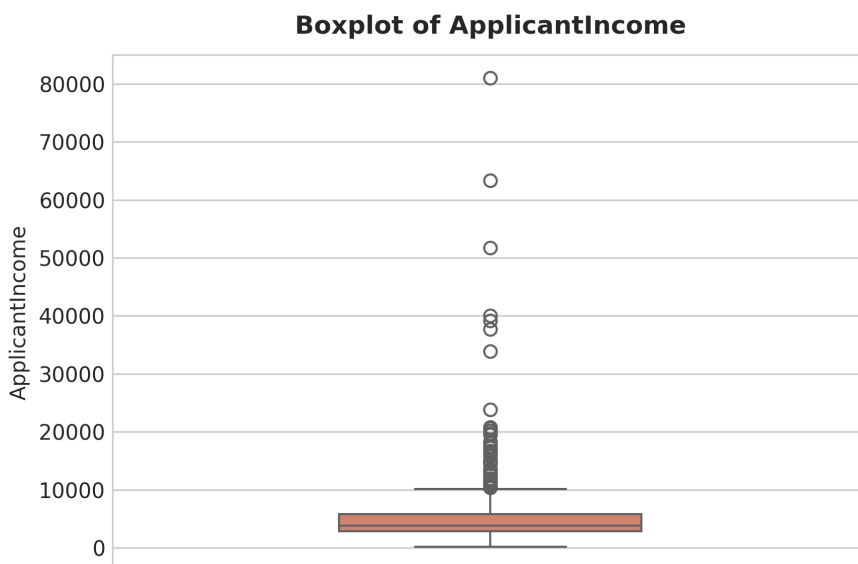


Figure 6: Box Plot of Applicant Income

Inference: The boxplot displays a compact interquartile range (IQR) with median income around \$3,812. Numerous upper outliers extend beyond the whisker bound, confirming high income dispersion.

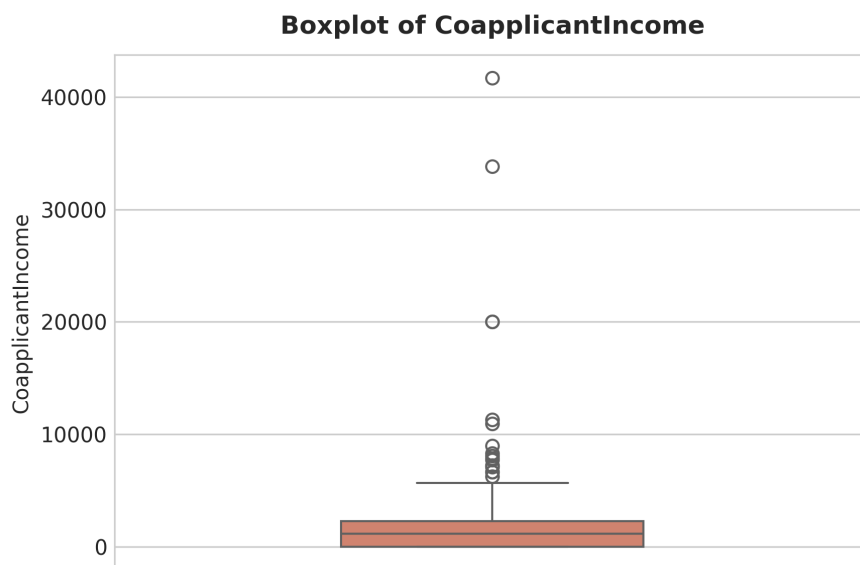


Figure 7: Box Plot of Co-applicant Income

Inference: Co-applicant income shows a low median near \$1,188, alongside multiple extreme values exceeding \$10,000.

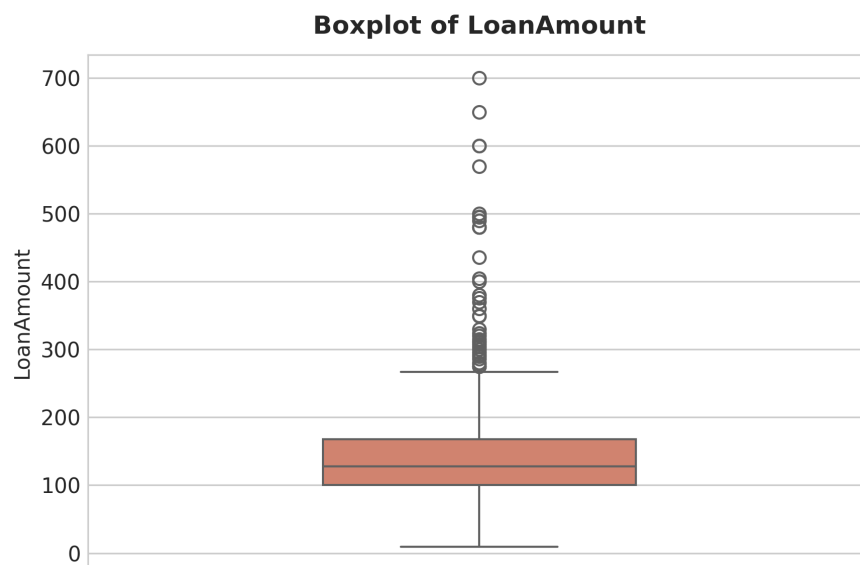


Figure 8: Box Plot of Loan Amount

Inference: The loan amount median sits near \$128,000. Outliers above \$260,000 highlight large loan applications requiring robust scaling during preprocessing.

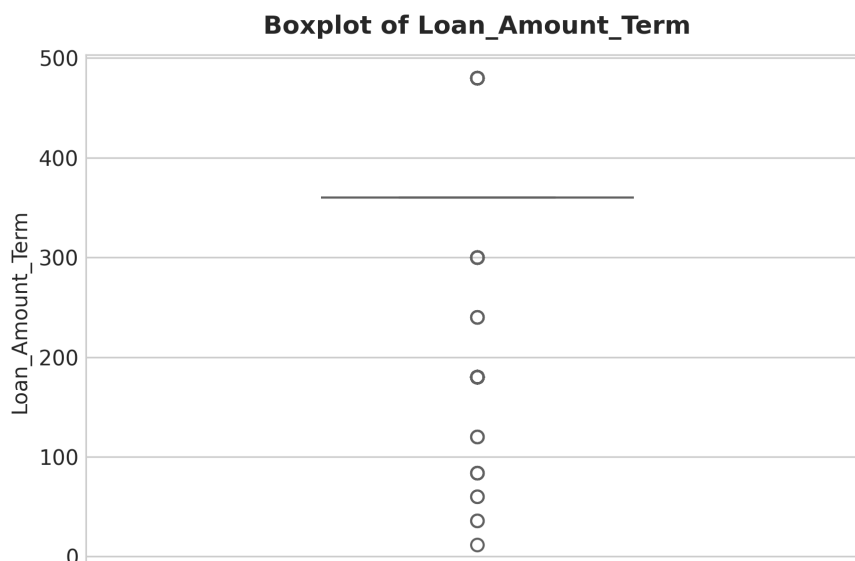


Figure 9: Box Plot of Loan Amount Term

Inference: The term variable is concentrated at 360 months. Non-standard loan durations (12 to 240 months and 480 months) appear as statistical outliers in the plot.

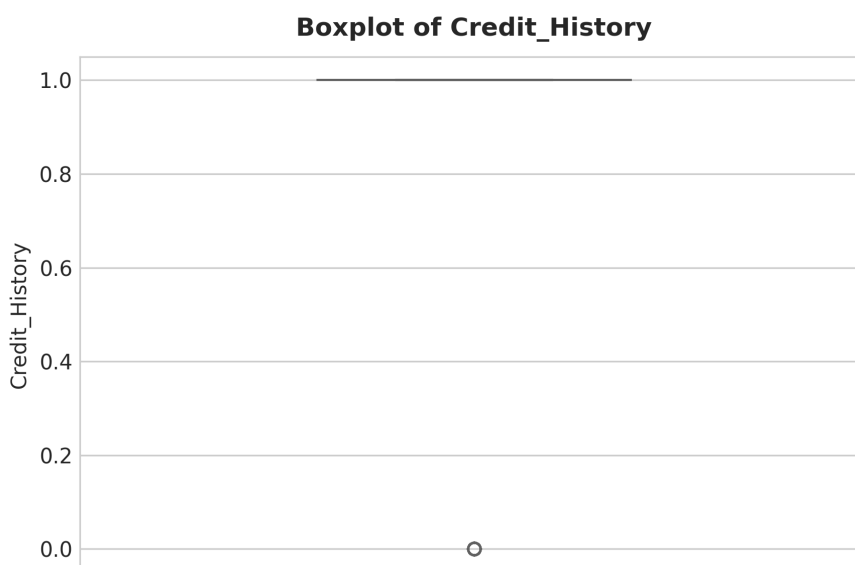


Figure 10: Box Plot of Credit History

Inference: The box plot compresses at 1.0 due to the high proportion of positive credit scores, reflecting discrete binary behavior.

4.3 Kernel Density Estimation (KDE) Plots

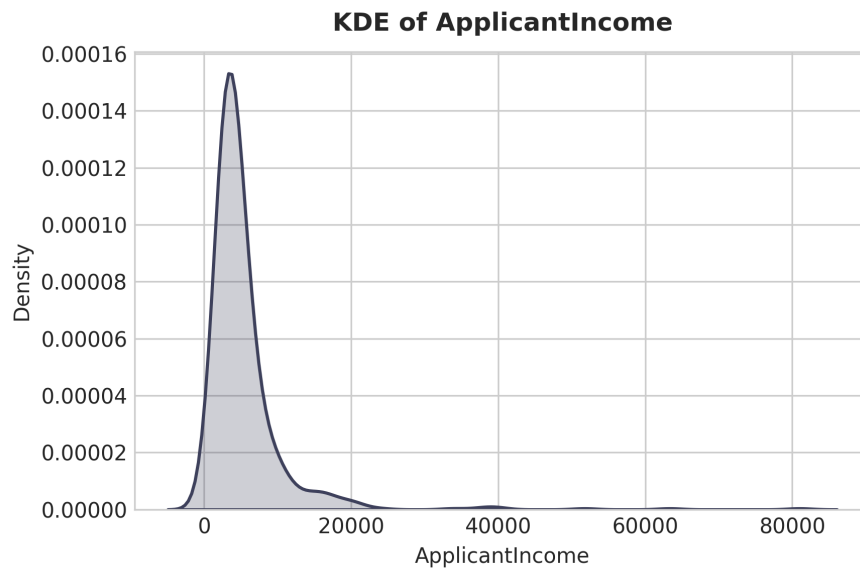


Figure 11: KDE of Applicant Income

Inference: The Gaussian smooth density curve highlights the single primary peak around \$3,500 with a long exponential decay toward higher income values.

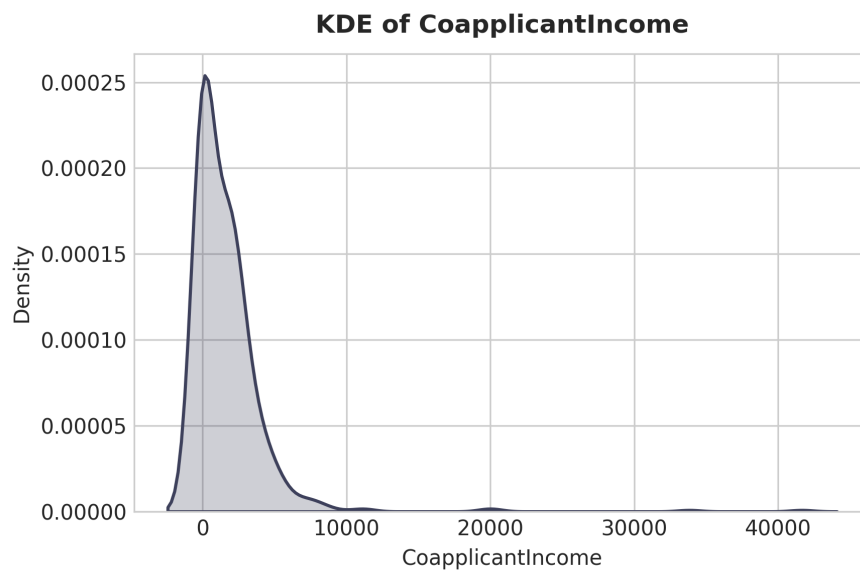


Figure 12: KDE of Co-applicant Income

Inference: Confirms a dominant density peak near \$0 representing solo applications, followed by a smoothly decaying distribution.

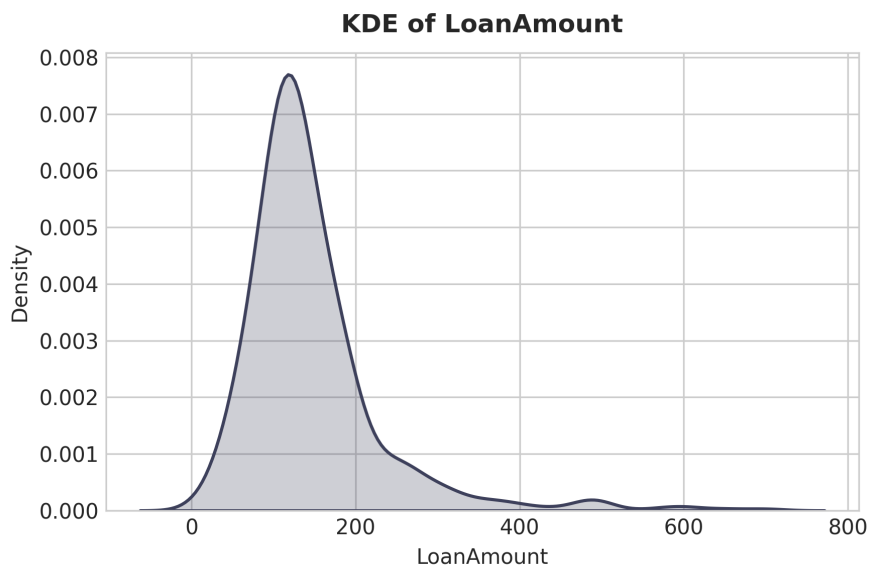


Figure 13: KDE of Loan Amount

Inference: The continuous density estimate displays near log-normal behavior, peaking cleanly around \$125,000.

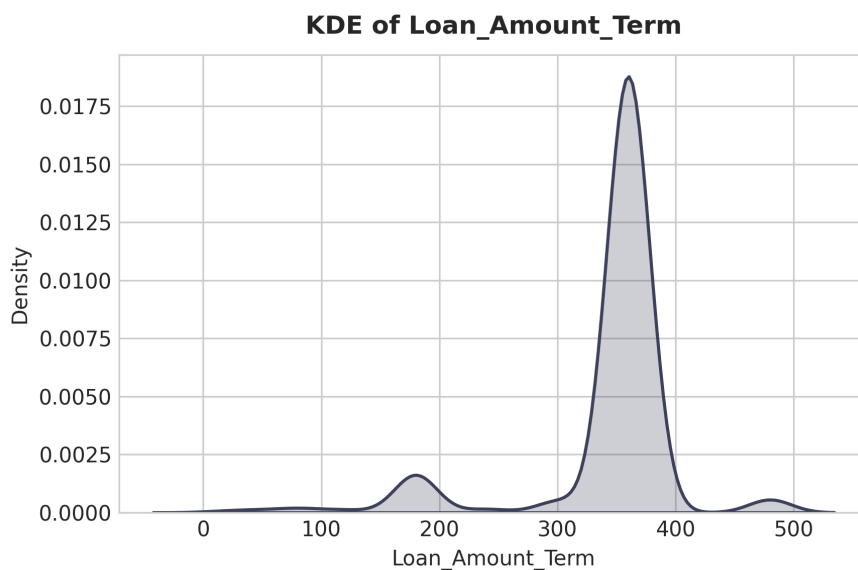


Figure 14: KDE of Loan Amount Term

Inference: Displays an extremely narrow peak centered precisely at 360 months.

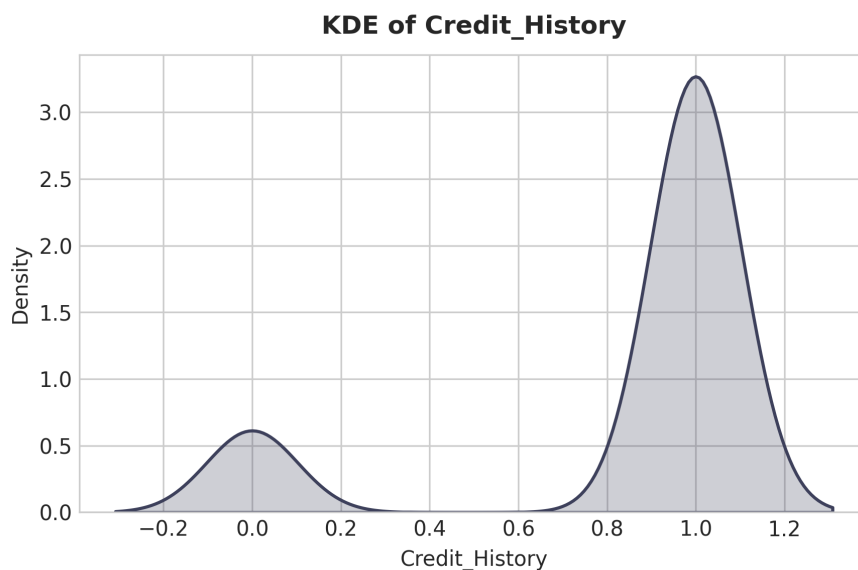


Figure 15: KDE of Credit History

Inference: Bimodal density peaks near 0.0 and 1.0 clearly illustrate the binary nature of historical credit standing.

4.4 Count Plots of Categorical Features

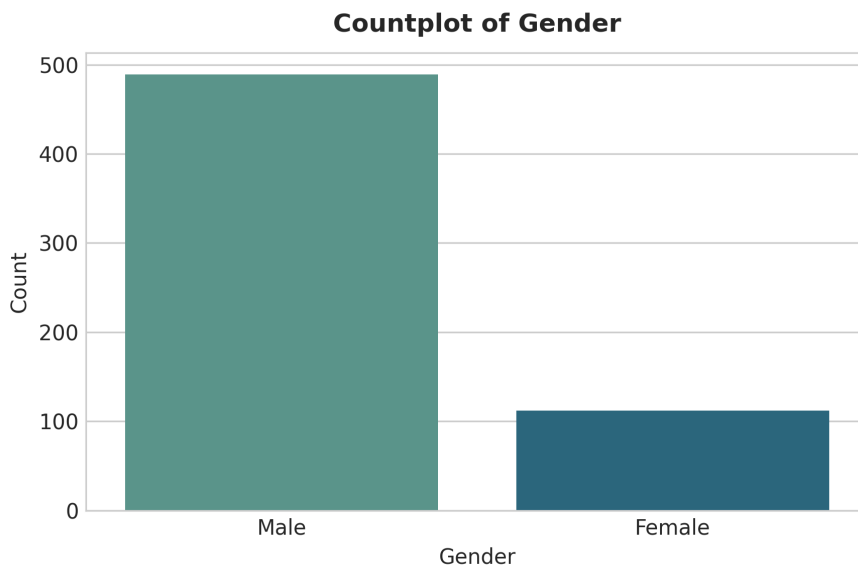


Figure 16: Count Plot of Gender

Inference: Male applicants comprise over 80% of the records, highlighting gender imbalance within the historical sampling pool.

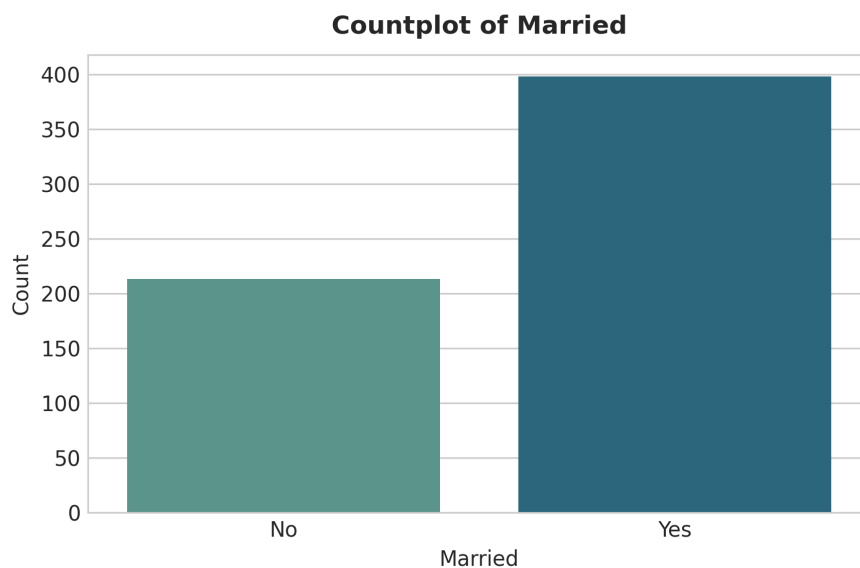


Figure 17: Count Plot of Married

Inference: Married individuals account for roughly two-thirds of total applications, reflecting joint credit applications.

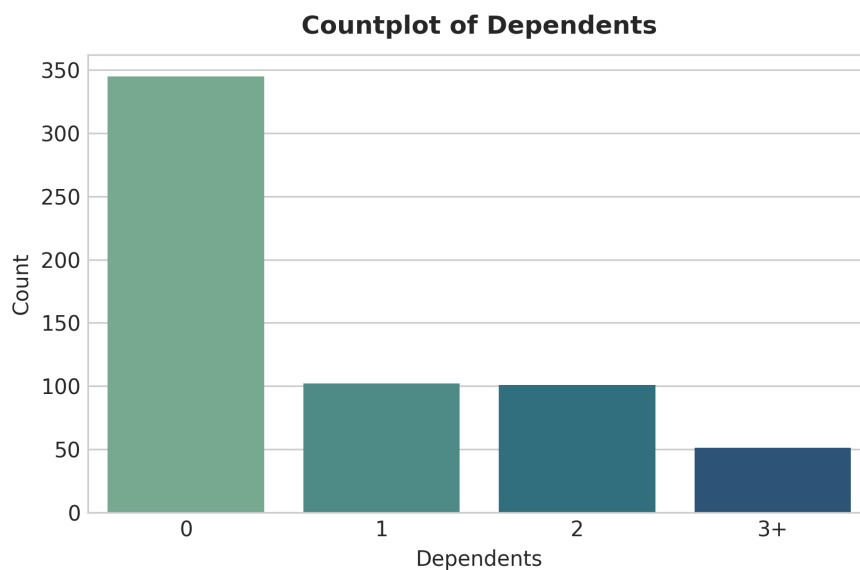


Figure 18: Count Plot of Dependents

Inference: Applicants with 0 dependents form the largest demographic group, followed by 1, 2, and 3+ dependents in descending frequency.

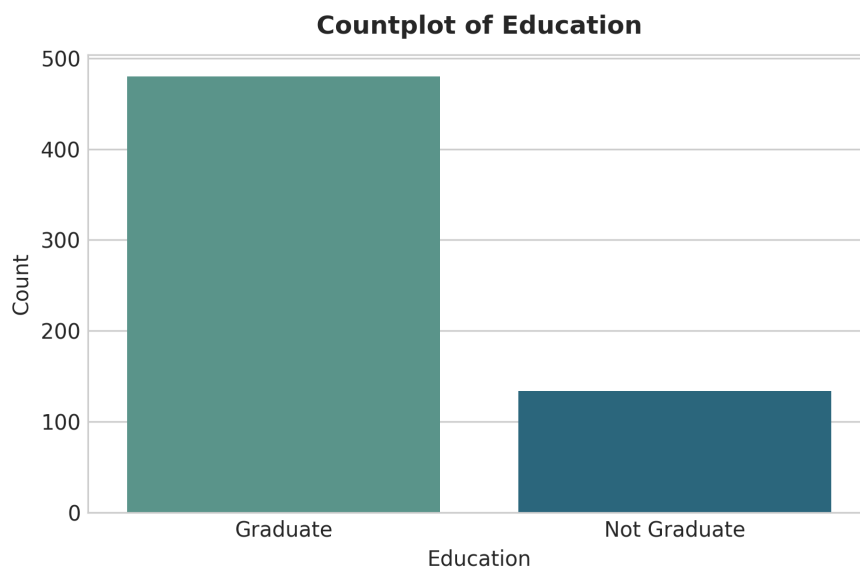


Figure 19: Count Plot of Education

Inference: Graduates constitute approximately 78% of total applicants compared to non-graduates.

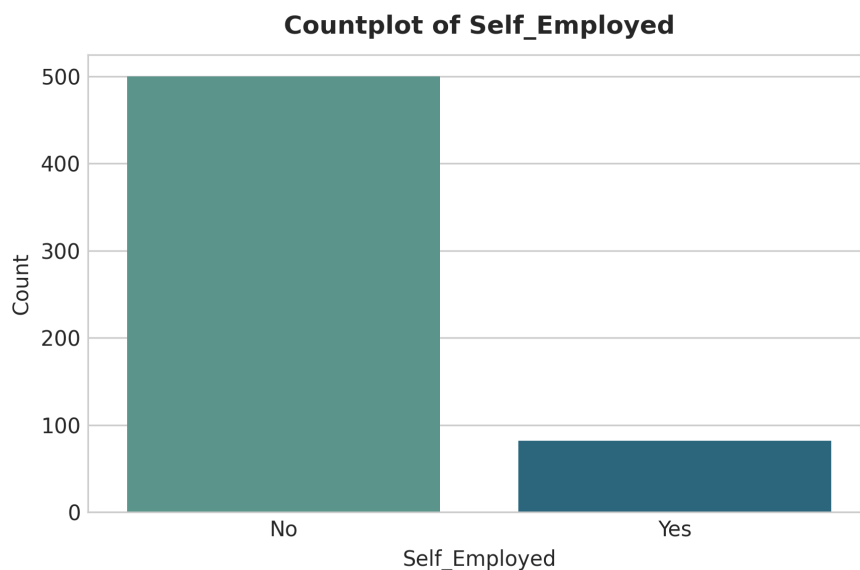


Figure 20: Count Plot of Self Employed

Inference: Self-employed individuals represent only 14% of applicants, with the majority being salaried employees.

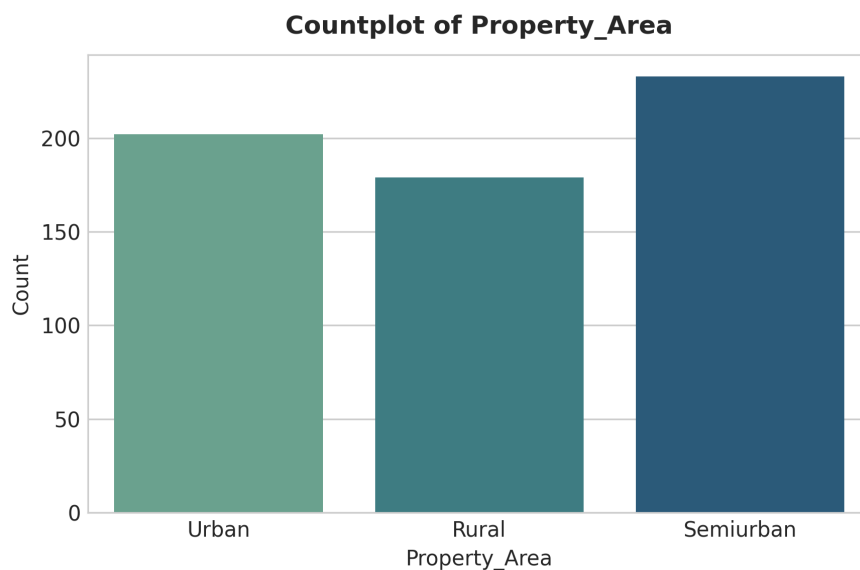


Figure 21: Count Plot of Property Area

Inference: Property locations are fairly balanced across categories, with Semiurban slightly exceeding Urban and Rural areas.

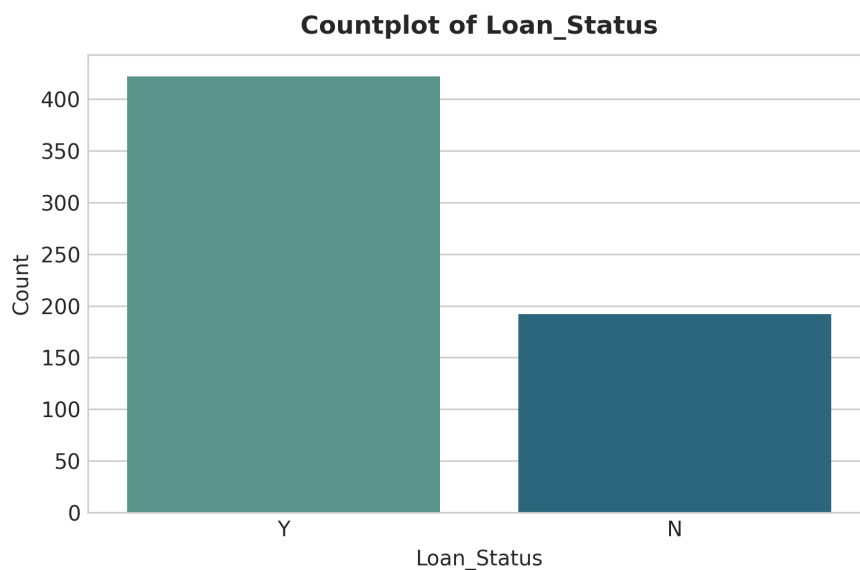


Figure 22: Count Plot of Loan Status

Inference: Approximately 69% of loan applications in the dataset were approved (Y), while 31% were rejected (N).

4.5 Correlation Heatmap of Numerical Features

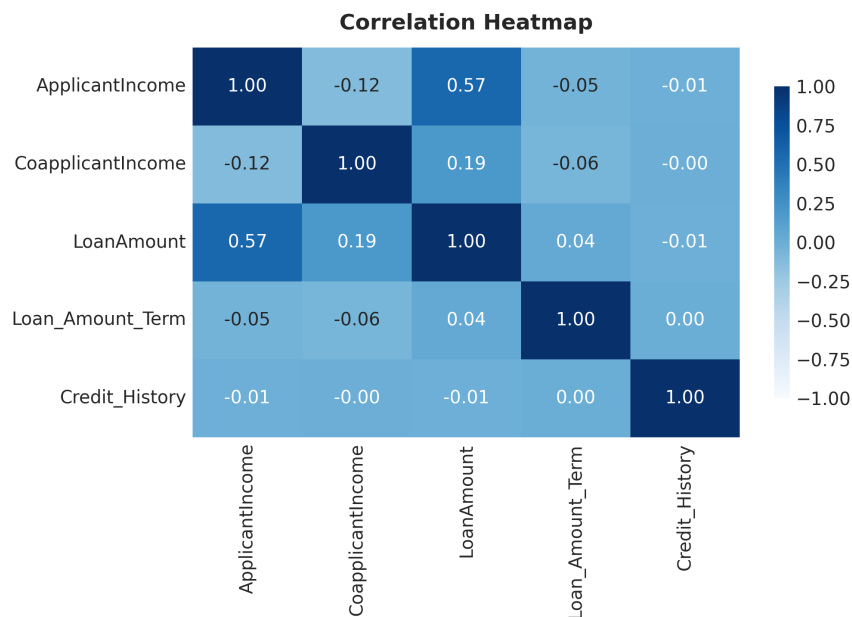


Figure 23: Correlation Heatmap

Inference: A moderate positive linear correlation ($r = 0.57$) exists between **ApplicantIncome** and **LoanAmount**, demonstrating that higher earnings translate to larger requested loans. Other numerical pairs show weak linear dependencies.

4.6 Pairwise Scatter Matrix (Pair Plot)

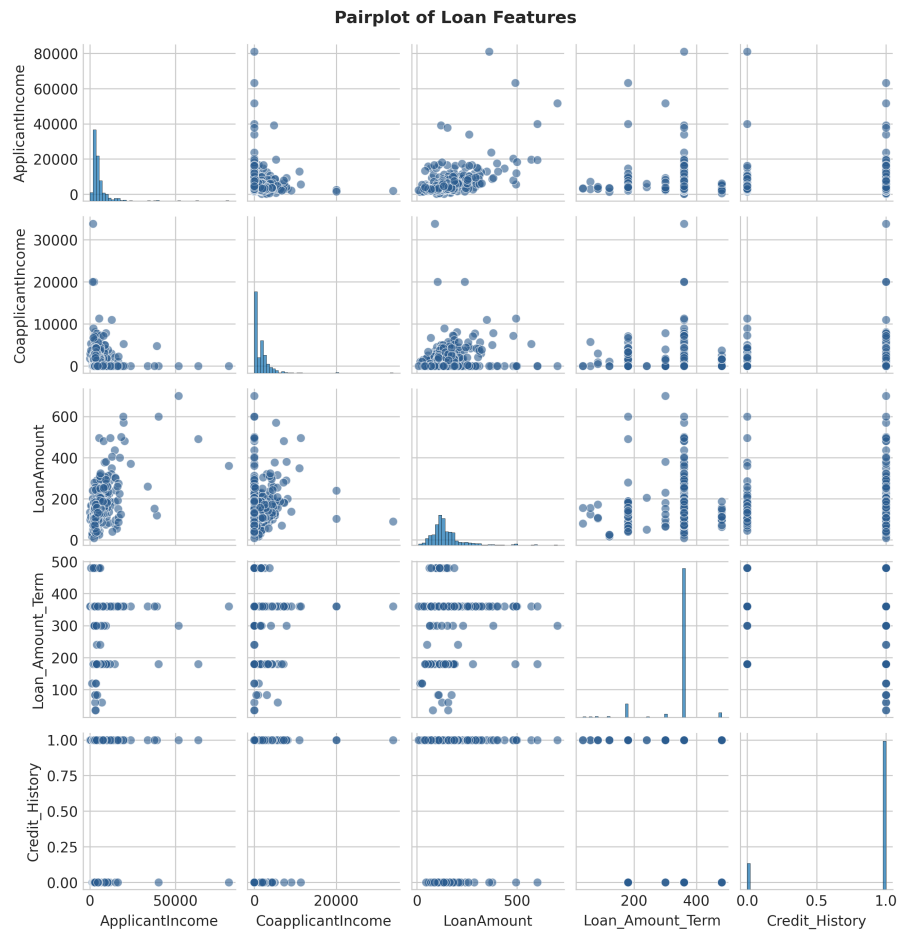


Figure 24: Pair Plot of Features

Inference: The scatter matrix highlights bivariate feature interactions alongside diagonal marginal distributions. The upward linear trend between primary income and loan amount is evident across the scatter grid.

4.7 Bivariate Scatter Plots

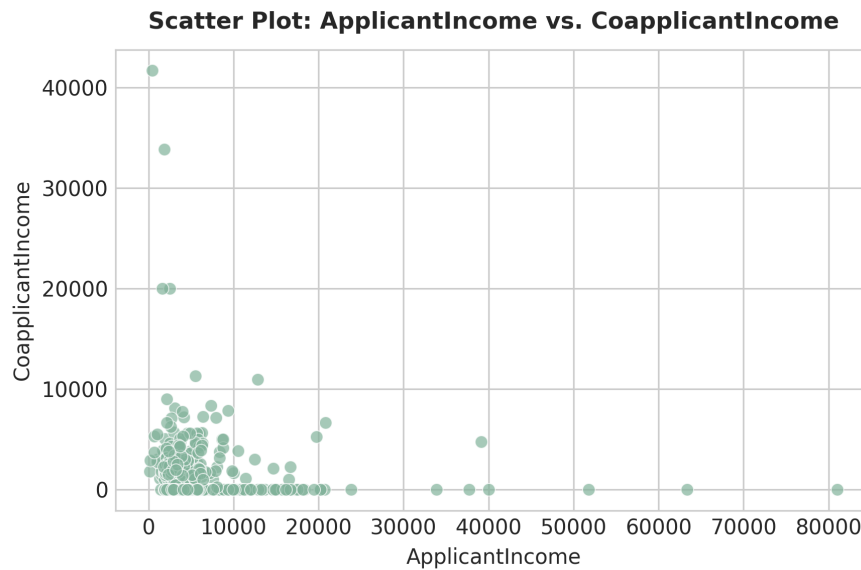


Figure 25: Scatter Plot: Applicant Income vs. Co-applicant Income

Inference: Data points cluster near the origin, indicating no linear dependency between primary and secondary applicant incomes.

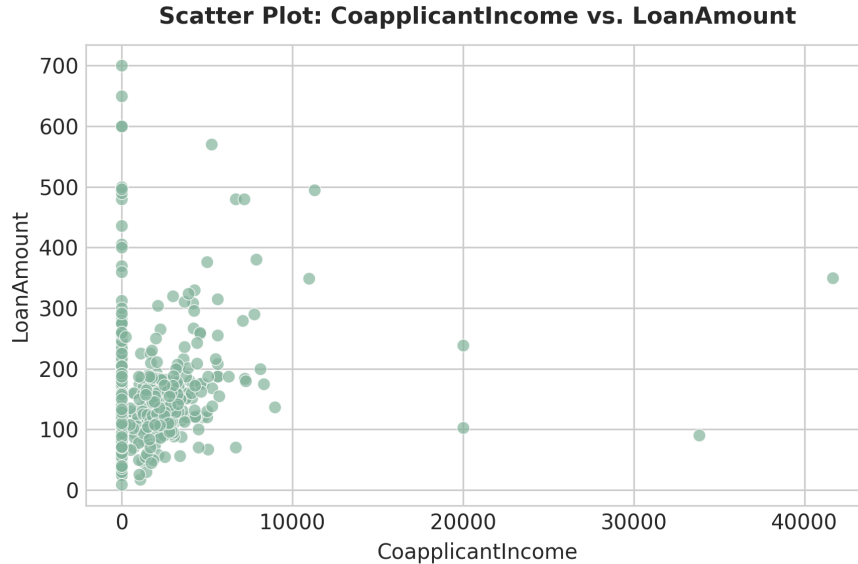


Figure 26: Scatter Plot: Co-applicant Income vs. Loan Amount

Inference: Displays a weak positive association ($r = 0.19$) with substantial dispersion across income levels.

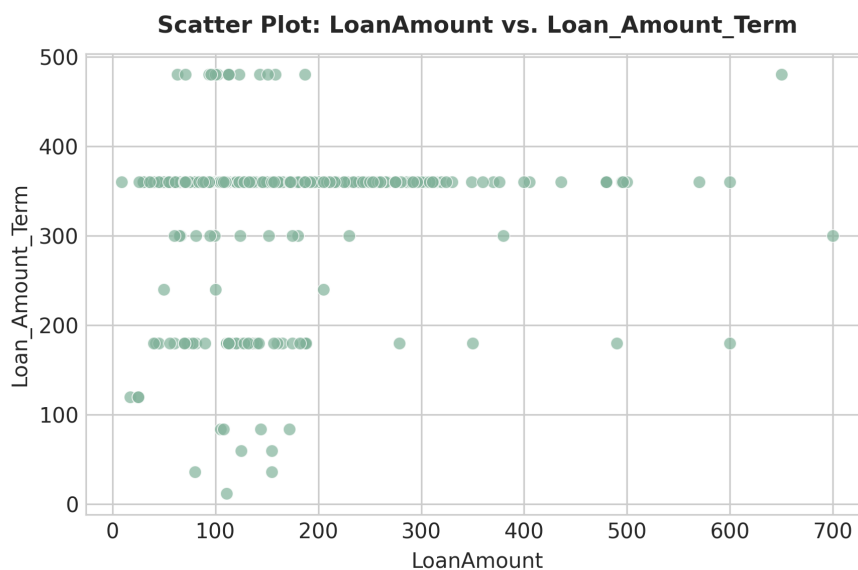


Figure 27: Scatter Plot: Loan Amount vs. Loan Amount Term

Inference: Points align along discrete horizontal bands corresponding to standard term lengths (e.g. 180 and 360 months).

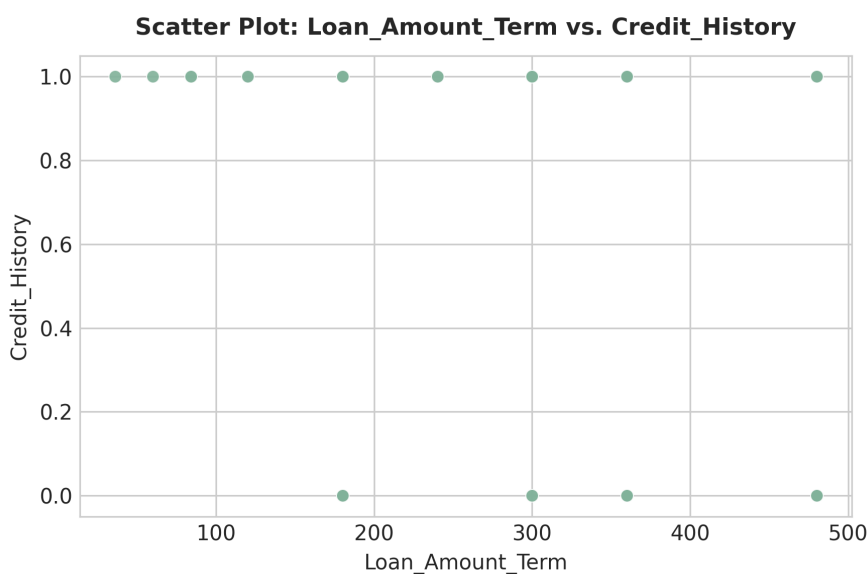


Figure 28: Scatter Plot: Loan Amount Term vs. Credit History

Inference: Bivariate points are localized at discrete grid intersections, reflecting the categorical nature of both variables.

5 Data Preprocessing

The automated EDA framework handles essential preprocessing checks:

- **Type Profiling:** Automatically partitions columns into numerical, categorical, and text types.
- **Null Detection:** Identifies missing values across key columns (`Gender`, `Married`, `Dependents`, `Self_Employed`, `LoanAmount`, `Loan_Amount_Term`, `Credit_History`). Recommended resolution involves median imputation for numerical data and mode imputation for categorical attributes.
- **Duplicate Verification:** Checks for duplicate rows across samples (0 duplicates found).
- **Standardization:** Recommends z-score scaling ($z = \frac{x-\mu}{\sigma}$) prior to fitting distance-based models.

6 Machine Learning Algorithms

The table below categorizes the five benchmark datasets explored in this experiment according to task type, feature selection, and algorithm choices.

Table 2: Benchmark Dataset Mapping and Model Selection

Dataset Name	Type of Task	ML	Feature Selection	ML Algorithm
Iris Dataset	Supervised Multiclass Classification		SelectKBest (ANOVA F-test) / PCA	Decision Trees, Support Vector Machines, KNN
Loan Amount Prediction	Supervised Regression		Correlation Analysis / Mutual Information	Linear Regression, Random Forest Regressor
Predicting Diabetes	Supervised Binary Classification		Chi-Square Test / SelectKBest	Logistic Regression, Support Vector Classifier
Classification of Email Spam	Supervised Binary Classification		Information Gain / TF-IDF Vectorization	Multinomial Naïve Bayes, SVM
Handwritten Character / MNIST	Supervised Multiclass Classification		Principal Component Analysis (PCA)	Convolutional Neural Networks (CNN), SVM

7 Experimental Setup

All automated scripts and visualization modules were executed under the setup detailed below.

Component	Environment Specification
Python Environment	Version 3.10+
Pandas Library	Version 2.0+
NumPy Library	Version 1.24+
Matplotlib / Seaborn	Matplotlib 3.7+ / Seaborn 0.12+
Execution Interface	Jupyter Notebook / Integrated Terminal
Operating System	Windows 11

8 Hyperparameter Selection

As this experiment focuses on automated exploratory analysis, hyperparameter tuning for predictive models was omitted. Visualization parameters were set as follows:

- **KDE Bandwidth:** Standard Gaussian kernel density estimator with automatic bandwidth calculation.
- **Histogram Binning:** Bins dynamically assigned based on the Freedman-Diaconis rule.
- **Correlation Matrix:** Pearson correlation coefficient (r) computed across numerical feature pairs.

9 Results

The automated pipeline extracted summary statistics across all numerical features in the Loan dataset.

Table 3: Loan Dataset Descriptive Summary Statistics

Metric	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term
Mean	\$5,403.46	\$1,621.24	\$146.41k	342.00 months
Std Dev	\$6,109.04	\$2,926.25	\$85.59k	65.12 months
Minimum	\$150.00	\$0.00	\$9.00k	12.00 months
Median (50%)	\$3,812.50	\$1,188.50	\$128.00k	360.00 months
Maximum	\$81,000.00	\$41,667.00	\$700.00k	480.00 months

10 Result Visualization

All 28 visualization plots were exported cleanly to `figures_loan/`. The analysis confirmed zero duplicate rows, identified skewness across financial features, and demonstrated a positive linear relationship ($r = 0.57$) between applicant income and requested credit amount.

11 Discussion

Automated exploratory tools provide fast, reproducible, and standardized data quality insights prior to model development.

- **Advantages:** Instantly surfaces missing values, feature skewness, and class imbalance without requiring manual plotting code.
- **Limitations:** Pairwise plotting scales quadratically ($\mathcal{O}(d^2)$) with feature count, which can increase memory overhead for high-dimensional data.
- **Future Extensions:** Incorporate interactive HTML dashboard generation (e.g. via Sweetviz or ydata-profiling) and automated outlier flags.

12 Conclusion

This experiment established an automated EDA workflow using core Python packages (NumPy, SciPy, Pandas, Matplotlib, Seaborn, and Scikit-Learn). Applied to the Loan dataset, the pipeline generated 28 diagnostic graphics revealing feature distributions, correlation patterns, and preprocessing needs. Additionally, five benchmark datasets were categorized into appropriate machine learning workflows, providing a solid foundation for model implementation.

13 References

- [1] Pandas Development Team, “Pandas API Reference,” <https://pandas.pydata.org/docs/reference/index.html>.
- [2] NumPy Developers, “NumPy User Guide,” <https://numpy.org/doc/stable/user/index.html>.
- [3] Seaborn Project, “Seaborn: Statistical Data Visualization,” <https://seaborn.pydata.org/>.
- [4] UCI Machine Learning Repository, “Loan Approval Dataset,” <https://archive.ics.uci.edu/ml/index.php>.